

Penetration Testing & Scan Report

 **CRITICAL EXPOSURE ASSESSMENT — TARGET HOST 10.0.2.3**

Target IP	10.0.2.3	Scan Timestamp	2026-08-03 10:09 (+0530)
MAC Address	08:00:27:92:E3:48	NIC Vendor	Oracle VirtualBox
Host Status	UP (0.0011s latency)	Assessor Workstation	adil@kali

1. EXECUTIVE SUMMARY

An active network reconnaissance scan was executed using Nmap 7.99 against target host **10.0.2.3**. The scan uncovered a **critically vulnerable posture**. Unlike standard hardened environments, this system exposes **5 open TCP ports** running severely outdated and end-of-life software services without any firewall filtering. OS fingerprinting identifies the system as running an outdated **Linux 2.6.x kernel**, matching the signature of a vulnerable test system (such as Metasploitable2).

2. REMOTE OPERATING SYSTEM FINGERPRINT (-O)

CRITICAL RISK SYSTEM DETECTED: The remote target is running an ancient **Linux 2.6.X** kernel branch (Linux 2.6.17 - 2.6.36). This kernel is known to contain multiple unpatched Local Privilege Escalation (LPE) vulnerabilities (e.g., Dirty COW, Kernel Ring Buffer exploits).

PARAMETER	OBSERVED TECHNICAL DATA
Device Type	General Purpose Server
Running Kernel	Linux 2.6.X
OS Fingerprint Match	Linux 2.6.17 – 2.6.36
CPE Identifier	cpe:/o:linux:linux_kernel:2.6
Network Distance	1 hop (Local Virtual Subnet)

3. PORT & SERVICE IDENTIFICATION (-SS, -SU, -SV)

PORT/PROTO	STATE	SERVICE	SERVICE VERSION / BANNER DETAILS	RISK LEVEL
21 / TCP	OPEN	ftp	ProFTPD 1.3.2c	CRITICAL
22 / TCP	OPEN	ssh	OpenSSH 5.3p1 (Debian 3ubuntu4 / Ubuntu Linux)	HIGH
80 / TCP	OPEN	http	Apache httpd 2.2.14 (DAV/2 mod_ssl/2.2.14 OpenSSL/0.9.8l PHP/5.3.1 mod_apreq2 mod_perl/2.0.4)	CRITICAL
443 / TCP	OPEN	ssl/http		CRITICAL

PORT/ PROTO	STATE	SERVICE	SERVICE VERSION / BANNER DETAILS	RISK LEVEL
Apache httpd 2.2.14 (Identical web stack as Port 80)				
3306 / TCP	OPEN	mysql	MySQL (unauthorized)	HIGH
53 / UDP	CLOSED	domain	—	NONE
67 / UDP	CLOSED	dhcps	—	NONE
161 / UDP	CLOSED	snmp	—	NONE

4. HIGH-RISK VULNERABILITY ANALYSIS

1. ProFTPD 1.3.2c Backdoor Command Execution (Port 21)

ProFTPD 1.3.2c is known to contain a malicious backdoor injected into its official source code archive (CVE-2010-4221 / OSVDB-69562). Attackers sending a specific command string can obtain immediate interactive command shell privileges on the host.

2. Outdated Web Application Stack & PHP 5.3.1 (Ports 80 / 443)

The target host runs Apache 2.2.14, PHP 5.3.1, and OpenSSL 0.9.8l. PHP 5.3.1 contains multiple remote code execution (RCE) flaws, while OpenSSL 0.9.8l is vulnerable to MITM renegotiation attacks and memory corruption exploits.

3. Exposed Database Service (Port 3306 - MySQL)

MySQL returns an `unauthorized` banner directly over the open network interface. This indicates active database listening without IP restrictions, enabling remote brute-force attacks and protocol exploits.

5. REMEDIATION RECOMMENDATIONS

- **Isolate Target:** Instantly isolate target host `10.0.2.3` from any non-isolated network segments.
- **Re-image Server OS:** Upgrade the operating system from Linux 2.6.x to a supported, modern distribution (e.g., Ubuntu 22.04 LTS / Debian 12).
- **Patch Software Services:** Update Apache to 2.4.x+, PHP to 8.x, OpenSSH to modern versions, and eliminate ProFTPD 1.3.2c immediately.
- **Enforce Firewall Controls:** Configure UFW/iptables to block public access to administrative ports like MySQL (3306).